

Python - Booleans

Python Booleans (bool)

In Python, **bool** is a sub-type of **int** type. A bool object has two possible values, and it is initialized with Python keywords, **True** and **False**.

Example

```
>>> a=True
>>> b=False
>>> type(a), type(b)
(<class 'bool'>, <class 'bool'>)
```

A bool object is accepted as argument to **type conversion** functions. With True as argument, the `int()` function returns 1, `float()` returns 1.0; whereas for False, they return 0 and 0.0 respectively. We have a one argument version of **complex()** function.

If the argument is a complex object, it is taken as real part, setting the imaginary coefficient to 0.

Example

```
a=int(True)
print ("bool to int:", a)
a=float(False)
print ("bool to float:", a)
```

```
a=complex(True)
print ("bool to complex:", a)
```

On running this code, you will get the following **output** –

```
bool to int: 1
bool to float: 0.0
bool to complex: (1+0j)
```

Python Boolean Expression

Python boolean expression is an expression that evaluates to a Boolean value. It almost always involves a **comparison operator**. In the below example we will see how the comparison operators can give us the Boolean values. The bool() method is used to return the truth value of an expression.

```
Syntax: bool([x])
Returns True if X evaluates to true else false.
Without parameters it returns false.
```

Below we have examples which use numbers streams and Boolean values as parameters to the bool function. The results come out as true or false depending on the parameter.

Example

```
# Check true
a = True
print(bool(a))
# Check false
a = False
print(bool(a))
# Check 0
a = 0.0
print(bool(a))
# Check 1
a = 1.0
print(bool(a))
# Check Equality
```

```
a = 5
b = 10
print(bool( a==b))
# Check None
a = None
print(bool(a))
# Check an empty sequence
a = ()
print(bool(a))
# Check an empty mapping
a = {}
print(bool(a))
# Check a non empty string
a = 'Tutorialspoint'
print(bool(a))
```

TOP TUTORIALS

[Python Tutorial](#)

[Java Tutorial](#)

[C++ Tutorial](#)

[C Programming Tutorial](#)

[C# Tutorial](#)

[PHP Tutorial](#)



Chapters ▾

Categories ≡

[Python Tutorial](#)

[CSS Tutorial](#)

[JavaScript Tutorial](#)

[SQL Tutorial](#)

TRENDING TECHNOLOGIES

[Cloud Computing Tutorial](#)

[Amazon Web Services Tutorial](#)

[Microsoft Azure Tutorial](#)

[Git Tutorial](#)

[Ethical Hacking Tutorial](#)

[Docker Tutorial](#)
[Kubernetes Tutorial](#)
[DSA Tutorial](#)
[Spring Boot Tutorial](#)
[SDLC Tutorial](#)
[Unix Tutorial](#)

CERTIFICATIONS

[Business Analytics Certification](#)
[Java & Spring Boot Advanced Certification](#)
[Data Science Advanced Certification](#)
[Cloud Computing And DevOps](#)
[Advanced Certification In Business Analytics](#)
[Artificial Intelligence And Machine Learning](#)
[DevOps Certification](#)
[Game Development Certification](#)
[Front-End Developer Certification](#)
[AWS Certification Training](#)
[Python Programming Certification](#)

COMPILERS & EDITORS

[Online Java Compiler](#)
[Online Python Compiler](#)
[Online Go Compiler](#)
[Online C Compiler](#)
[Online C++ Compiler](#)
[Online C# Compiler](#)
[Online PHP Compiler](#)
[Online MATLAB Compiler](#)
[Online Bash Terminal](#)
[Online SQL Compiler](#)
[Online Html Editor](#)

[ABOUT US](#) | [OUR TEAM](#) | [CAREERS](#) | [JOBS](#) | [CONTACT US](#) | [TERMS OF USE](#) |
[PRIVACY POLICY](#) | [REFUND POLICY](#) | [COOKIES POLICY](#) | [FAQ'S](#)



Tutorials Point is a leading Ed Tech company striving to provide the best learning material on technical and non-technical subjects.

© Copyright 2026. All Rights Reserved.